

UPSC

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1. (a)

Average Price of Soap in 2013.Nos. of Soap ^{Sold} ~~Sales~~ by X Company

$$= \frac{3.5 \times 10^5}{7}$$

$$= 50,000$$

Nos. of Soap ^{Sold} ~~Sales~~ by Y Company.

$$= \frac{4.5 \times 10^5}{5}$$

$$= 90,000$$

Nos. of Soap ^{Sold} ~~Sales~~ by Z Company

$$= \frac{4 \times 10^5}{8}$$

$$= 50,000$$

Therefore,

Average Price of Soap in 2013

$$= \frac{\text{Total Price of all Soaps Sold by all Companies.}}{\text{Total nos. of unit sold}}$$

$$= \frac{350000 + 450000 + 400000}{50000 + 90,000 + 50000}$$

$$= \frac{1200000}{190000} = 6.315 / \text{unit}$$

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(5)

$$\begin{aligned}\text{Unit Sold for X in 2011} &= \frac{3 \times 10^5}{5} \\ &= \underline{60,000}\end{aligned}$$

$$\begin{aligned}\text{Unit Sold for X in 2012} &= \frac{3 \times 10^5}{6} \\ &= \underline{50,000}\end{aligned}$$

~~Change~~

Decrease in unit sold for X in 2012
over 2011

$$\begin{aligned}&= 60,000 - 50,000 \\ &= \boxed{10,000}\end{aligned}$$

$$\begin{aligned}\% \text{ Decrease} &= \frac{\text{Decrease in unit sold for X in 2012 over 2011}}{\text{Unit sold for X in 2011}} \times 100 \\ &= \frac{10,000}{60,000} \times 100 \\ &= \boxed{16.67\%}\end{aligned}$$

Hence,

% Decrease in unit sold for X in
2012 over 2011 is $\boxed{16.67\%}$

UPSC

Page No. :

Roll No. :

Name :

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$$\begin{aligned}\text{Unit sold for X in 2013} &= \frac{3.5 \times 10^5}{7} \\ &= \underline{50,000}\end{aligned}$$

$$\begin{aligned}\text{Unit sold for Y in 2013} &= \frac{4.5 \times 10^5}{5} \\ &= \underline{90,000}\end{aligned}$$

$$\begin{aligned}\text{Unit sold for Z in 2013} &= \frac{4 \times 10^5}{8} \\ &= \underline{50,000}\end{aligned}$$

$$\begin{aligned}\% \text{ share of Z} &= \frac{\text{Unit sold for Z in 2013}}{\text{total unit sold for X, Y \& Z in 2013}} \times 100 \\ &= \frac{50,000}{50,000 + 90,000 + 50,000} \times 100 \\ &= \boxed{26.315\%}\end{aligned}$$

Hence,

% share of Z in total units sold
in 2013 is $\boxed{26.315\%}$.

UPSC

Page No. :

Roll No. :

Name :

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(d)

$$\begin{aligned}\text{Unit Sold by Y in 2011} &= \frac{4.4 \times 10^5}{4.5} \\ &= \frac{440000}{4.5} = \boxed{110000}\end{aligned}$$

$$\begin{aligned}\text{Unit Sold by Y in 2012} &= \frac{4.2 \times 10^5}{6} \\ &= \boxed{70000}\end{aligned}$$

$$\begin{aligned}\text{Unit Sold by Y in 2013} &= \frac{4.5 \times 10^5}{5} \\ &= \boxed{90000}\end{aligned}$$

$$\begin{aligned}\text{Unit Sold by Y in 2014} &= \frac{4.9 \times 10^5}{7} \\ &= \boxed{70000}\end{aligned}$$

$$\begin{aligned}\text{Unit Sold by Y in 2015} &= \frac{4.2 \times 10^5}{6} \\ &= \boxed{70000}\end{aligned}$$

Hence,

from above assessment Unit Sold by
Y in year 2013 is maximum $\boxed{\text{i.e. } 90000}$

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(e)

Maximum price per unit is 9 ₹/unitMinimum price per unit is 4 ₹/unit

$$\begin{aligned}
 \text{Exceeds of maximum price over} \\
 \text{minimum price} &= \text{Maximum price} \\
 &\quad - \text{Minimum price} \\
 &= 9 - 4 \\
 &= \underline{5 \text{ ₹/unit}}
 \end{aligned}$$

$$\begin{aligned}
 \% \text{ Exceeds} &= \frac{\text{Exceeds of maximum price over minimum price}}{\text{Minimum price per unit}} \times 100 \\
 &= \frac{5}{4} \times 100 \\
 &= \boxed{125\%}
 \end{aligned}$$

Hence, Maximum price per unit given in the table exceeds the minimum price per unit by 125%.